Large-scale traffic flow reconstruction from heterogeneous mobility sensors

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Abstract

Mobile-phone data, traffic counters and GPS traces give complementary and incomplete views of road traffic. We combine them to reconstruct hourly flows on the regional road network of Emilia-Romagna, Italy. Mobile-phone origin–destination activity drives an agent-based simulation on an OpenStreetMap graph; inductive loop detectors correct the simulated counts locally, and the correction is propagated to neighbouring links with simulated turning shares. Viasat GPS probes are kept out of the reconstruction and used for validation after a link-specific penetration rescaling. On all usable matched links, 848 in a February 2020 week and 845 in a June–July 2021 week, the median absolute error of the hourly traffic shares is 1.34 and 1.49 percentage points, 32% and 36% of the mean hourly share, and 27% on the links with penetration near 1%. The simulated peaks precede the observed ones by about one hour. The reconstruction improves only slightly on the hourly profile of the injected demand (median error 1.40 and 1.52 points) and is less accurate than the mean profile of the regional loop detectors (1.15 and 1.21 points). In a leave-one-out test on 149 detectors, assimilating the others lowers the error on the held-out detector from 59.8% to 56.7% of its observed volume. Used as an exposure denominator for the 2022–2024 accident tables, the reconstructed field gives a relative risk that peaks around midnight and an accident share that scales across hours as $E^{0.72}$ (95% interval 0.58–0.88). The workflow runs as a webserver that delivers the assimilated link flows.

Keywords: traffic reconstruction, mobile-phone data, traffic counters, GPS telemetry, agent-based model, data fusion

1 Introduction

Passive mobility data have changed the scale at which urban and regional transport systems can be observed. This shift is part of a broader movement toward urban computing, smart-city analytics and digital representations of urban systems [1–5]. Mobile-phone records ~~reveal robust~~ show regular spatial and temporal ~~regularities~~ patterns in human movement [6, 7] and have been used to estimate origin–destination (OD) matrices for transport analysis [8–10]. More generally, large geolocated data sets can support travel-demand estimation and road-use analysis when they are combined with network information and independent validation ~~[11, 13].~~ [11–13]. **[TODO: cite earlier Bologna-group papers on Emilia-Romagna mobility; none are in the repository]** These data streams do not, however, remove the central inverse problem of traffic reconstruction: demand is observed at population or areal scale, whereas congestion, emissions and infrastructure assessment depend on flows on individual road links [14–16].

A link-level traffic field is also ~~a natural~~ an exposure denominator for ~~road safety~~ road-safety analysis. Crash-frequency and before–after studies ~~have long treated~~ use traffic volume or vehicle distance travelled ~~as a necessary reference for interpreting~~ to interpret accident counts, ~~because~~ since more accidents on a road or in an hour ~~can reflect greater exposure rather than intrinsically higher hazard~~ may simply reflect more vehicles [17–20]. Speed, traffic regime and road environment ~~further modulate this relationship [21, 22]. This makes reconstructed traffic useful beyond mobility description: once validated, it can be reused to~~ modulate this relation [21, 22]. A reconstructed field can normalize aggregate accident data ~~, detect temporal risk regimes and screen road segments whose hazard-weighted exposure is unusually high. The purpose is not to predict individual crashes, but to provide an interpretable exposure layer for regional road-safety prioritization~~in time and weight the simulated traffic of each link by the hours in which it occurs.

Each observational layer has a different failure mode. Mobile-phone OD data provide broad coverage but do not identify the exact road sequence followed by a trip, and trip-distribution models are known to be sensitive to spatial scale and distance-decay assumptions [23–26]. Fixed traffic counters measure vehicle passages directly, yet only at instrumented locations; using them to correct an OD matrix is a classical inverse problem with many more unknowns than counts [27]. Floating-car or GPS telemetry is spatially richer, but it measures the vehicles present in the provider's fleet and therefore requires a road-dependent penetration normalization before comparison with total traffic ~~[28, 30, 31]~~[28–31]. Treating these layers as interchangeable counts can produce non-physical aggregate validations, for example by summing flows on roads that do not represent the same traffic stream. The same issue appears more generally in heterogeneous urban data, where missing values, sampling mechanisms and data governance shape what can be inferred [32–34].

~~This paper proposes an operational~~ We build a link-level reconstruction workflow ~~that preserves the role of~~ in which each data source ~~. The contribution is not a new route-choice model, but a physically interpretable multi-layer reconstruction framework. The novelty is the integration of areal mobile-phone demand, a routable road graph, sparse fixed counters and independent GPS telemetry into a single decision-support chain while keeping their measurement meanings distinct~~keeps its own role. Mobile-phone OD activity ~~is used to generate~~ generates time-dependent demand. A directed road graph built from OpenStreetMap [35, 36] ~~provides the routing support, using~~ supports shortest-path ~~logic derived from Dijkstra's algorithm [37] and~~ routing [37] in the traffic-assignment tradition [14, 15]. ~~Magnetic coils locally assimilate~~ Inductive loop detectors of the regional traffic monitoring system (MTS), which we call coils below, correct the simulated link counts where ~~direct observations exist~~they are installed. Viasat GPS telemetry ~~is kept outside the reconstruction and data-assimilation steps; it~~ stays outside reconstruction and assimilation and is used only ~~as an independent validation layer for the simulation framework after estimating~~ for validation, after a link-specific penetration coefficient ~~, so that the comparison evaluates temporal-profile consistency rather than absolute GPS fleet size . The validation links are then clustered using Ward hierarchical~~ removes the size of the probe fleet. We validate the hourly profiles on all usable matched links of a 2020 and a 2021 week, compare them with two baseline profiles, test the assimilation by leaving out one coil at a time, and use Ward clustering [38, 39] to ~~distinguish different temporal road~~ choose example links with different daily profiles. The same ~~reconstructed field is~~

exposed through a dashboard/webserver so that simulation runs, data-assimilated flows, monitoring views and exports can be used by analysts without reproducing the full modelling chain in code. This operational layer is important for digital-twin field is served by a webserver to analysts who do not run the modelling chain [2, 5], and smart-city applications, where reconstructed network fields must be inspected, shared and reused for downstream questions such as traffic management and safety screening [2, 5, 40?]we use it as an exposure layer for aggregate accident data.

The manuscript is organized as follows. Results and validation first present the end-to-end workflow, from Section 2 presents the data layersto the representative validation links, then show how the reconstructed traffic field can be reused as an exposure layer for road-safety screening, and finally describe the operational webserverbuilt around the same reconstruction chain. Methods then summarize the data, the `minimocas` traffic model, , the reconstruction, the coil-based data-assimilation step, accident exposure analysis, the validation against Viasat and the exposure-normalized risk analysis and the GPS webserver. Section 3 describes data, model, assimilation, risk indicators and validation procedure. Additional explanations of data-layer meaning, numerical parameters, validation thresholds, road-safety descriptors and webserver architecture are reported Section 4 discusses the results and their limitations, and Section 5 concludes. Tables and further details are in the Supplementary Material.

2 Results and validation

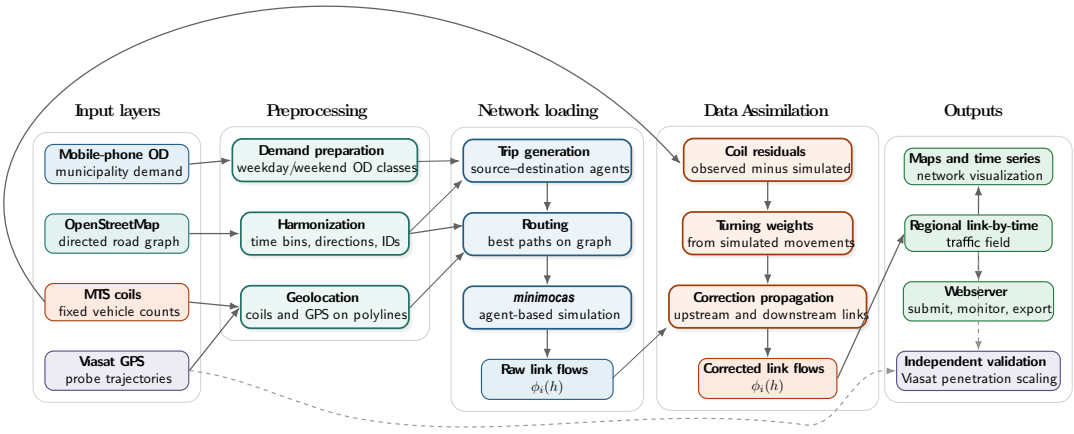


Fig. 1 End-to-end reconstruction workflow. Mobile-phone OD activity defines regional demand, OpenStreetMap provides the directed road graph, MTS coils are geolocated on road polylines and Viasat GPS traces are kept for independent validation of the simulation framework. Demand is injected into the `minimocas` agent-based simulator, producing raw link flows. Coil residuals are then used for local data assimilation of the simulated counts and propagated through adjacent directed links using simulated turning weights. The final product is a data-assimilated link-by-time traffic field that can be mapped, validated against held-out GPS traces and exposed through the webserver.

Figure 1 summarizes the logic of the reconstructionbefore the individual empirical results are presented. The workflow is intentionally modular: mobile-phone data define reconstruction. Mobile-phone data define the demand, the road graph defines admissible movementthe admissible movements, coils constrain local volumes through data assimilation, and GPS traces are held out for independent validationof the simulation framework. This separation makes the data-assimilation step interpretable because the information added by fixed counters is propagated through validation. Information from the fixed counters spreads along the simulated turning

structure~~rather than through a spatial smoothing rule unrelated to traffic direction~~ , so it follows the direction of traffic.

2.1 Observation layers and mobile-phone demand

The reconstruction is carried out on the Emilia-Romagna regional road network, whose main structure is an east–west corridor connected to dense urban areas, coastal roads and Apennine links (Fig. 2a). The mobile-phone layer is aggregated at municipality level and converted into OD activity. Weekday activity in January 2020 is strongly concentrated along the regional urban corridor, especially around Bologna and the main provincial centres (Fig. 2b). ~~This spatial heterogeneity is the demand signal that the model must redistribute~~ The model has to redistribute this demand onto individual road segments ~~, following the general idea that aggregate mobility observations must be coupled to an explicit spatial network before they can inform link-level transport questions~~ [8, 11, 15].

The ~~regional road traffic flow monitoring system (MTS)~~ MTS coils provide local vehicle counts on selected roads (Fig. 2c). Coil 282 is used as an illustrative fixed counter because it has a complete directional time series for the selected day and an unambiguous geographical record. Its 15-min profile ~~displays the expected daily structure of a road counter:~~ has low night traffic, a morning ~~increase and sustained afternoon activity~~ peak at 07:00 in direction 1 and an evening peak at 17:00 in direction 0 (Fig. 2d). ~~This combination of sparse~~ Sparse fixed detectors and broader but sampled mobile traces ~~is typical~~ are the typical inputs of traffic-state reconstruction ~~problems~~ [28, 31].

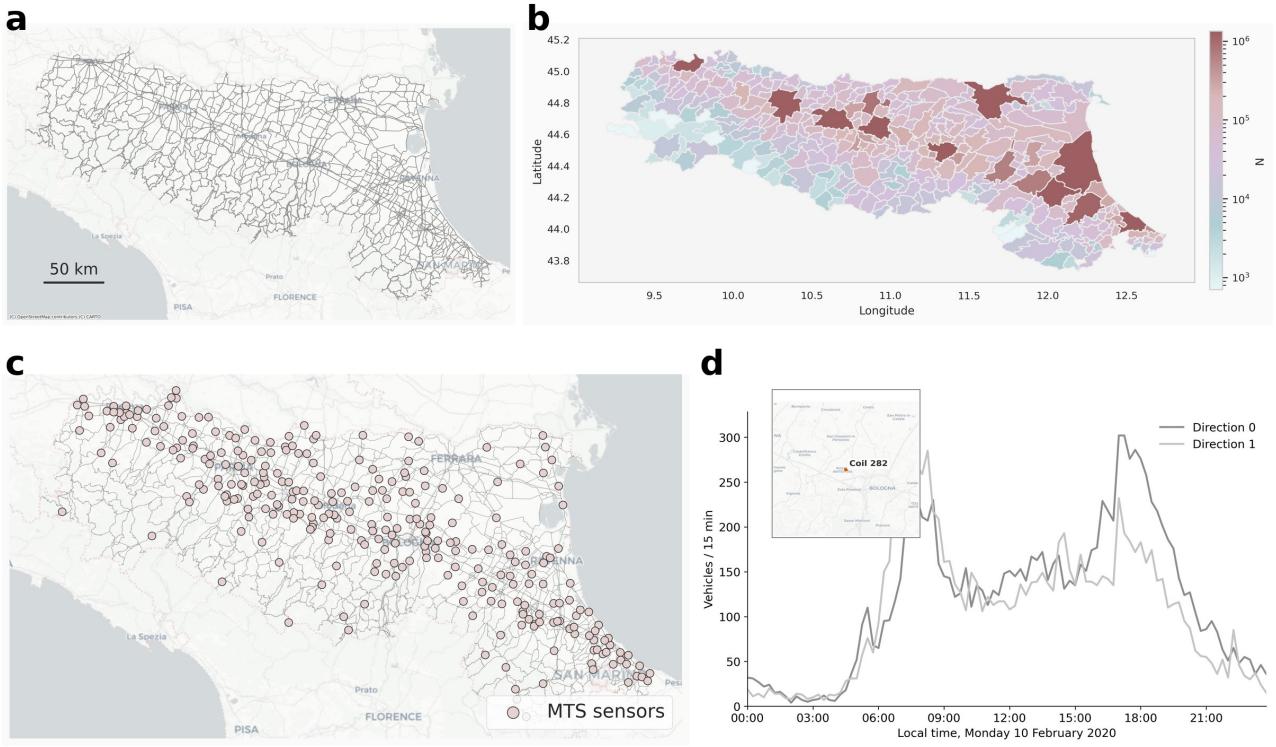


Fig. 2 Study domain and observational inputs. (a) Regional road cartography used by the model on a CartoDB Positron basemap; the scale bar corresponds to 50 km. **(b)** Weekday mobile-phone OD activity over Emilia-Romagna municipalities in January 2020; colour encodes activity N on a logarithmic scale. **(c)** ~~Magnetic traffic-system coil locations~~ Locations of the MTS inductive loop detectors (coils) over the regional road graph. **(d)** ~~Detailed location~~ Location of coil 282 (inset) and complete 15-min directional vehicle counts ~~for Wednesday~~on Monday 10 February 2020, ~~15 March 2020.~~local time.

The demand layer also contains temporal and categorical information. Mean weekday activity has a sharper morning peak and a stronger late-afternoon component than weekend activity (Fig. 3a). The trip-generation classes indicate whether both endpoints are observed in the OD data or whether one or both endpoints are sampled from the modelled area (see Methods, Section 3.1). Figure 3b,c shows the mobility components used to decompose agent behaviour from the OD data. In both weekdays and weekends, the source–destination component dominates the reconstructed demand, while random-endpoint components represent residual mobility that cannot be directly linked to two observed municipalities. ~~These classes are not arbitrary noise: they regularize~~ They spread weakly observed OD mass ~~, prevent sparse low-count pairs from generating deterministic routes with false precision, preserve regional coverage and avoid an artificial collapse of weakly observed mobility onto a small number of~~ over the network, so that it does not pile up on a few high-count municipality pairs. Ranking municipalities by activity confirms that a small number of urban centres drive a large fraction of the demand (Fig. 3d), consistent with empirical evidence that human mobility is highly heterogeneous across places and distances [6, 23, 24].

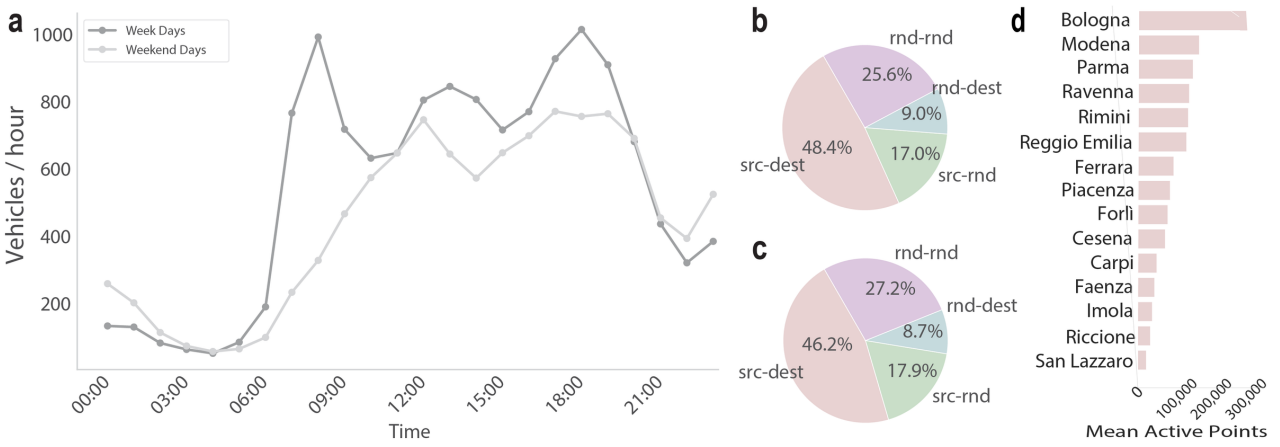


Fig. 3 Mobile-phone demand used to drive the reconstruction. (a) Mean hourly OD activity profiles for weekdays and weekend days in January 2020. **(b)** Weekday and **(c)** weekend shares of the four generation classes: source–destination (`src-dest`), source–random (`src-rnd`), random–destination (`rnd-dest`) and random–random (`rnd-rnd`). **(d)** ~~Ranking~~ Mean mobile activity of the 15 most active municipalities~~by mean mobile activity; the plot reports the first 10 and last 10 municipalities among the top 50.~~.

2.2 Network reconstruction and local counter control

The simulation converts the OD demand into directional link flows. For each hour, generated agents are placed on the directed road graph, routed toward their destination and counted on every traversed polyline. The output is a link-by-time matrix $\phi_i(h)$, where i is a road polyline and h is the hour. ~~The comparison with coil 282 shows that the simulated road associated with the counter follows the same broad daily structure as the observed series on~~ On Monday 10 February 2020, a working day before the COVID-19 mobility restrictions, the raw simulated flow on the polyline of coil 282 follows the daily structure of the counter (hourly correlation 0.95) with 67% of its daily volume, 14,804 against 21,935 vehicles (Fig. 4a). After assimilation the link carries the coil series by construction; the one-hour offset between the two curves comes from the UTC clock of the coil table (Section 4.1). The map of simulated daily flows highlights the high-volume regional corridors while retaining non-zero information on the secondary network (Fig. 4b). This microscopic-to-link aggregation follows

the logic of agent-based traffic simulators, where individual vehicle trajectories are used to recover network-level flow fields [41–43].

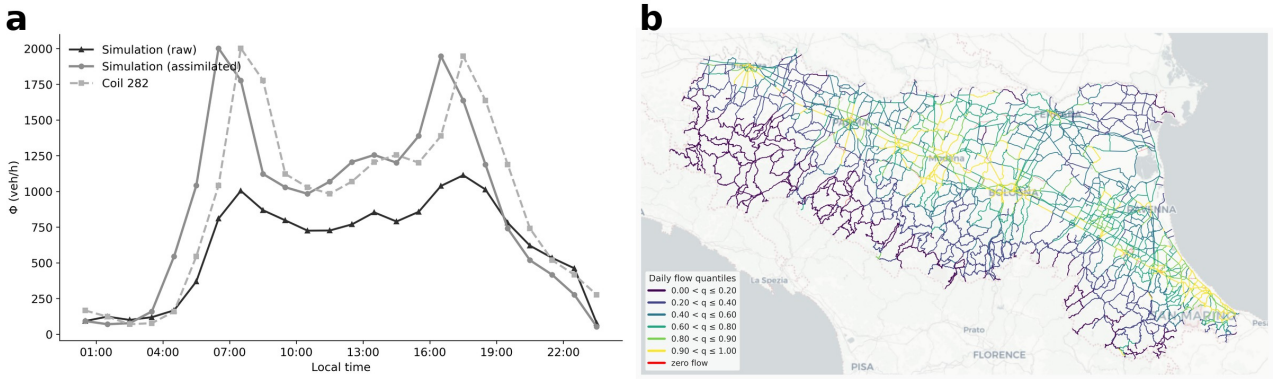


Fig. 4 Link-level reconstruction output. (a) ~~Hourly~~ Raw and assimilated simulated flow on the polyline ~~associated with~~ of coil 282 ~~compared with~~ and hourly ~~aggregated~~ coil ~~observations~~ counts on ~~15 March~~ Monday 10 February 2020, local time; Φ denotes vehicle flow. **(b)** Spatial distribution of simulated daily link flows over Emilia-Romagna, expressed through flow quantile classes; yellow segments carry the highest reconstructed flows and purple segments the lowest.

2.3 ~~Accident exposure and emergent road-risk structure~~

~~The reconstructed link-by-time traffic field is primarily a traffic product. As a downstream proof of concept, it also provides a dynamic exposure layer that can be reused for regional road-safety screening. Accident records available for Emilia-Romagna in 2022–2024 are aggregated and are not geolocated at the level of single events. For this reason, the analysis does not attempt to validate point hotspots or infer event-level causality. Instead, it asks whether the temporal and spatial structure of simulated exposure can organize aggregate accident information in a way that is useful for exposure normalization and prioritization.~~ The assimilation corrects the coil links themselves; how far the correction reaches is measured by a leave-one-out test on 10–17 February 2020. For each of the 149 coils with complete data on at least one day (610 coil-days), we remove the coil from the assimilation and from the set of polylines that stop the propagation, rerun the assimilation with the other coils, and compare the hourly counts on its link with the observation (Supplementary Section S4). Summed over all coil-days, the absolute error on the held-out link is 59.8% of the observed volume for the raw simulation and 56.7% after assimilation; the median over coils goes from 46.2% to 44.3%. The propagation from neighbouring coils reaches 51% of the held-out links, and the error decreases on 39% of the coils and increases on 13%. On the Viasat links the median profile error defined in Section 2.5 is 1.33 pp before and 1.34 pp after assimilation (Table 2). The assimilation therefore anchors the field at the instrumented sections and has a small effect a few links away from them.

~~We first compare the hourly accident distribution with simulated exposure profiles for weekdays and holidays (Fig. 5a)~~

2.3 Accident exposure

The reconstructed field can serve as an exposure denominator for aggregate accident statistics. The regional accident tables for 2022–2024 (16,679, 16,808 and 16,757 accidents) give counts by hour of the day and by road type, with no location. Let $A(h)$ be the share of ~~daily accidents observed~~ accidents in hour h ~~, and let~~ and $E(h)$

~~be the corresponding~~ the share of simulated ~~exposure. The exposure-normalized relative risk is defined as~~

$$R(h) = \frac{A(h)}{E(h)}.$$

vehicles on the whole network in the same hour (Fig. 5a). We define the relative risk

$$R(h) = \frac{A(h)/E(h)}{\frac{1}{24}\sum_{h'} A(h')/E(h')}, \tag{1}$$

~~This normalization separates hours with many accidents because many vehicles are present from hours where the accident share is high relative to the simulated exposure.The resulting profile shows higher relative risk during night-time hours and morning transitions, while central and evening periods with more stable traffic conditions tend to fall below the daily-average risk~~ which equals one on average over the day. With weekday exposure, R is 3.4 at 00:00 and 2.4 at 01:00, stays between 1.0 and 1.4 from 03:00 to 07:00, and is below 0.6 after 19:00 (Fig. 5b). ~~Clustering the hourly normalized-risk profiles identifies distinct temporal regimesrather than a single smooth daily curve~~ The night values rest on the simulated night exposure, the least reliable part of the simulated profile (Section 4.1). Ward clustering of the hourly risk and exposure values separates three regimes: night (21:00–06:00), morning (07:00–09:00) and day (10:00–20:00) (Fig. 5c).

~~Across hours, accident share scales sublinearly with simulated weekday exposure, $A \sim E^{\alpha}$, with $\alpha \simeq 0.72$ and $R^2 \simeq 0.85$~~ A least-squares fit of $\log A$ against $\log E$ on the 24 hourly points gives $A \sim E^{\alpha}$ with $\alpha = 0.72$, 95% bootstrap confidence interval 0.58–0.88 (10,000 resamples of the 24 hours), and $R^2 = 0.85$ (Fig. 5d). ~~The exponent should not be read as a causal parameter. It indicates that exposure explains a large part of the temporal organization of accidents, but that accidents do not increase in direct proportion to traffic volume. This is consistent with the idea that traffic amount, traffic regime, road environment and behavioural conditions jointly shape aggregate risk~~Across hours, the accident share grows less than proportionally with the simulated exposure.

~~The same risk profile can be projected back onto the simulated road network as~~ $R(h)$ is a regional profile and is the same for every link. The hazard-weighted exposure ~~(H_i~~ of a link (Eq. (12), Fig. 6a) ~~. This produces a link-level screening layer that highlights where large simulated flows coincide with hours of elevated relative risk. The weekday–holiday comparison emphasizes how the same physical network can carry different hazard-weighted exposure patterns under different demand regimes (~~is therefore its simulated volume weighted by the hourly composition of its traffic, and the rank difference S_i (Eq. (15), Fig. 6~~b) . Ranking links by their share of total hazard-weighted exposure defines a risk centrality ; the highest-ranked links are geolocated on the road map to make their spatial interpretation explicit~~ e) picks out links that carry a larger share of their traffic in high-R hours. The risk centrality C_i (Fig. 6c~~),and their relative shares are shown as a compact bar ranking~~ ,d) and the weekday–weekend contrast D_i (Fig. 6~~d) . Comparing hazard rank with exposure rank then identifies segments whose hazard is anomalously high relative to their simulated exposure (~~b) follow the

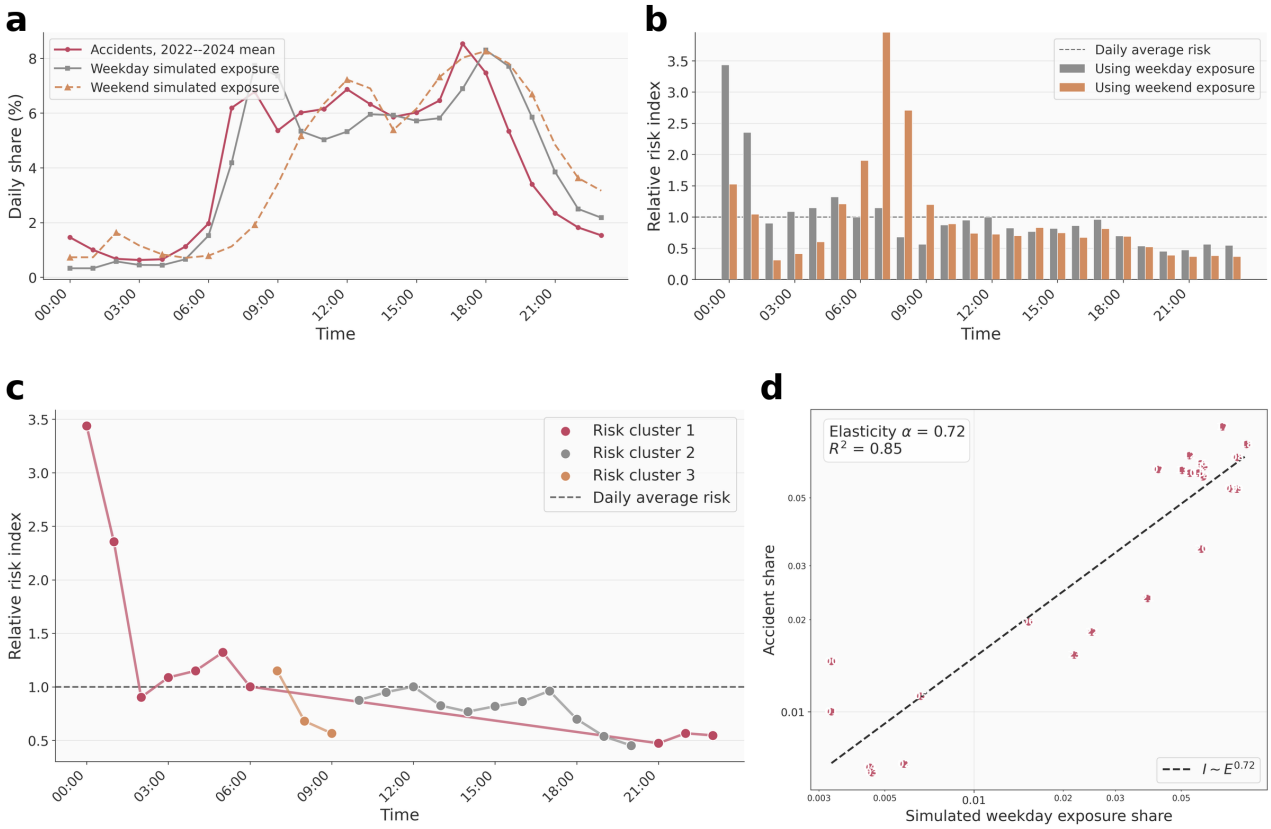


Fig. 5 ~~**Temporal risk/exposure analysis.**~~ **Accidents and simulated exposure by hour.** **(a)** ~~Mean hourly~~ Hourly accident share ~~for~~ (mean of 2022–2024~~compared with~~) and simulated weekday and ~~holiday~~ weekend exposure ~~profiles~~shares. **(b)** ~~Exposure-normalized hourly accident intensity for~~ Relative risk $R(h)$, Eq. (1), with weekday (grey) and ~~holiday~~ weekend (orange) exposure; the dashed line marks ~~the daily-average risk~~$R = 1$. **(c)** ~~Clustering of~~ Three hourly ~~normalized-risk profiles into three distinct temporal risk~~ regimes ~~; dashed line marks the daily-average~~ from Ward clustering of risk and exposure values. **(d)** ~~Scaling relation between~~ Accident share against simulated weekday exposure ~~and accident~~ share ~~across~~ for the 24 hours (log–log axes) and fit $A \sim E^{\alpha}$, ~~showing sublinear behaviour with elasticity~~ $\alpha = 0.72$ ~~and~~ (95% bootstrap interval 0.58–0.88), $R^2 = 0.85$.

same logic. These maps show where simulated traffic coincides with hours of high regional risk; they carry no information on where accidents occur. Supplementary Fig. 6~~e). These maps are intended for stakeholder prioritization and exploratory screening, not as evidence that specific accidents occurred on the highlighted links. Additional descriptive statistics on accident~~ S3 gives the yearly totals, road ~~environment~~ types and seasonal shares ~~are reported in Supplementary Fig. S1~~of the accident data.

2.4 Independent Viasat validation

Viasat GPS observations ~~provide~~ give an independent test of the ~~temporal~~ hourly profiles produced by the ~~simulation framework. Coverage is not uniform: matched Viasat links occupy~~ reconstruction; they enter neither the demand nor the assimilation. Matched Viasat links cover only part of the ~~complete~~ regional cartography, ~~with the validation focus~~ mostly in the Forlì–Cesena area (Fig. 7a,d). For each matched ~~road~~ polyline i we estimate ~~a statistical penetration coefficient~~ the penetration coefficient

$$\eta_i = \frac{\sum_h \psi_i(h)}{\sum_h \phi_i(h)} \frac{\sum_d \sum_h \psi_i(d,h)}{\sum_d \sum_h \phi_i(d,h)}, \tag{2}$$

where ~~$\psi_i(h)$ is the raw Viasat probe count and $\phi_i(h)$~~ $\psi_i(d,h)$ is the number of Viasat probes on day d and hour h, $\phi_i(d,h)$ is the simulated flow~~. The incremented GPS profile used for validation is~~ , and both sums run

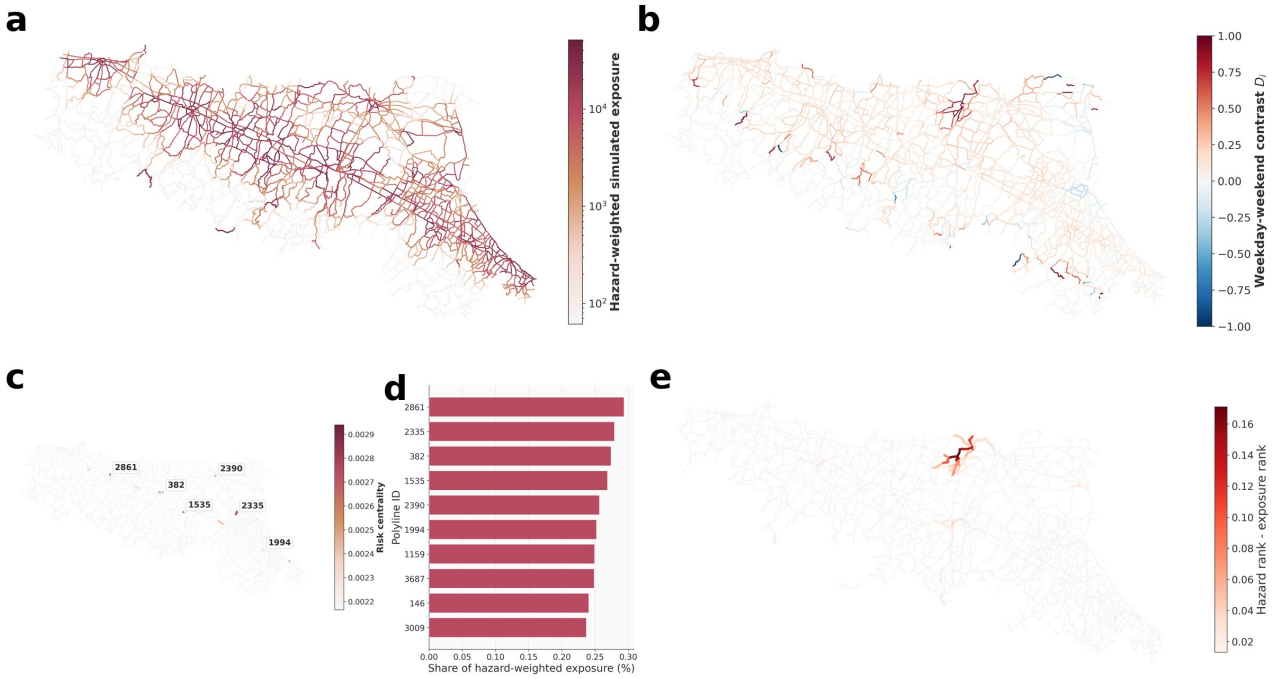


Fig. 6 ~~**Spatial organization of hazard-weighted simulated exposure.**~~ **Simulated weekday exposure weighted by the regional risk profile.** **(a)** Hazard-weighted exposure ~~on the simulated regional road network~~H_i, ~~obtained by weighting link-level simulated traffic with the hourly accident-risk profile~~Eq. (12). **(b)** ~~Relative weekday–holiday difference in hazard-weighted exposure~~Weekday–weekend contrast D_i, Eq. (13); positive values ~~indicate~~ mark links with ~~higher~~ larger weekday ~~hazard~~H_i. **(c)** ~~Geolocated highest-risk-centrality links; annotated polyline identifiers correspond to~~ Location of the ~~top-ranked segments visible on~~ links with the ~~map~~highest risk centrality C_i, Eq. (14), and **(d)** ~~Bar ranking of links by~~ their share of the total hazard-weighted exposure~~; risk centrality is the fraction of regional hazard-weighted traffic carried by each link~~. **(e)** ~~Road links with anomalously high hazard relative to their simulated exposure, identified by comparing hazard rank and exposure~~ Links in the top 5% of the rank ~~. Panels (a–e) represent simulated exposure weighted by aggregate temporal accident risk~~difference S_i, ~~not geolocated crash events~~Eq. (15).

over the same days of the validation interval. The penetration-normalized GPS profile is

$$\tilde{\psi}_i(h) = \frac{\psi_i(h)}{\eta_i} \frac{1}{\eta_i} \sum_d \psi_i(d,h). \tag{3}$$

~~This scaling aligns total volumes on each road and preserves~~ The rescaling sets the GPS volume of each road equal to the simulated one and leaves the hourly shape ~~of the GPS signal. Absolute GPS counts are not directly comparable with total simulated traffic because Viasat is partial-fleet telemetry and penetration varies spatially. The validation therefore evaluates whether the reconstructed daily profile is consistent with the independent probe profile after normalization, which isolates temporal consistency from fleet-size differences. It is not a validation of absolute fleet size , nor an estimate of a universal GPS market penetration. This~~ unchanged, so the comparison tests when traffic occurs on each link and says nothing about the size of the Viasat fleet. A link-specific ~~adjustment is important when probe-vehicle~~ factor is needed because probe coverage varies strongly across the network [28, 30, 31]~~,~~.

Validation compares the hourly shares $p_i(h) = \phi_i(h)/\sum_{h'} \phi_i(h')$ and ~~avoids treating raw GPS probe counts as complete traffic observations~~$q_i(h) = \tilde{\psi}_i(h)/\sum_{h'} \tilde{\psi}_i(h')$, where $\phi_i(h) = \sum_d \phi_i(d,h)$; the shares do not depend

on η_i. The profile error of link i is the mean absolute error over the 24 hours, in percentage points (pp),

$$\mathrm{MAE}_i = \frac{100}{24} \sum_{h=0}^{23} \left| p_i(h) - q_i(h) \right|, \tag{4}$$

and we also report it relative to the mean hourly share $100/24 \simeq 4.17$ pp,

$$\varepsilon_i = \frac{\mathrm{MAE}_i}{100/24} = \sum_{h=0}^{23} \left| p_i(h) - q_i(h) \right|, \tag{5}$$

which is the L^1 distance between the two daily distributions: $\varepsilon_i = 0$ for identical profiles and $\varepsilon_i = 2$ for profiles with disjoint support.

The penetration ~~distribution is broad~~spans four decades (Fig. 7b,e). In 2020 ~~, the three main penetration decades $\eta \sim 10^{-1}$, $\eta \sim 10^{-2}$ and $\eta \sim 10^{-3}$ contain 57.3%, 36.4% and 6.3% of classified roads, respectively (Fig. 7b)~~ the median η_i is 0.50%; the classes $\eta \sim 10^{-2}$ and $\eta \sim 10^{-3}$ contain 60.6% and 32.4% of the 848 usable links, and only 17 links (2.0%) reach $\eta \sim 10^{-1}$. In 2021 the ~~corresponding shares are 48.9% , 45.3% and 5.8% (Fig. 7c). The maps show that penetration classes are spatially interwoven , rather than forming a single compact sampling zone~~ median is 0.44% and the two main classes contain 61.3% and 34.2% of 845 links. The classes are interwoven in space (Fig. 7c,f)~~. This motivates a validation based on individual links and profile classes rather than on a single regional aggregate~~: penetration is a property of individual roads more than of an area.

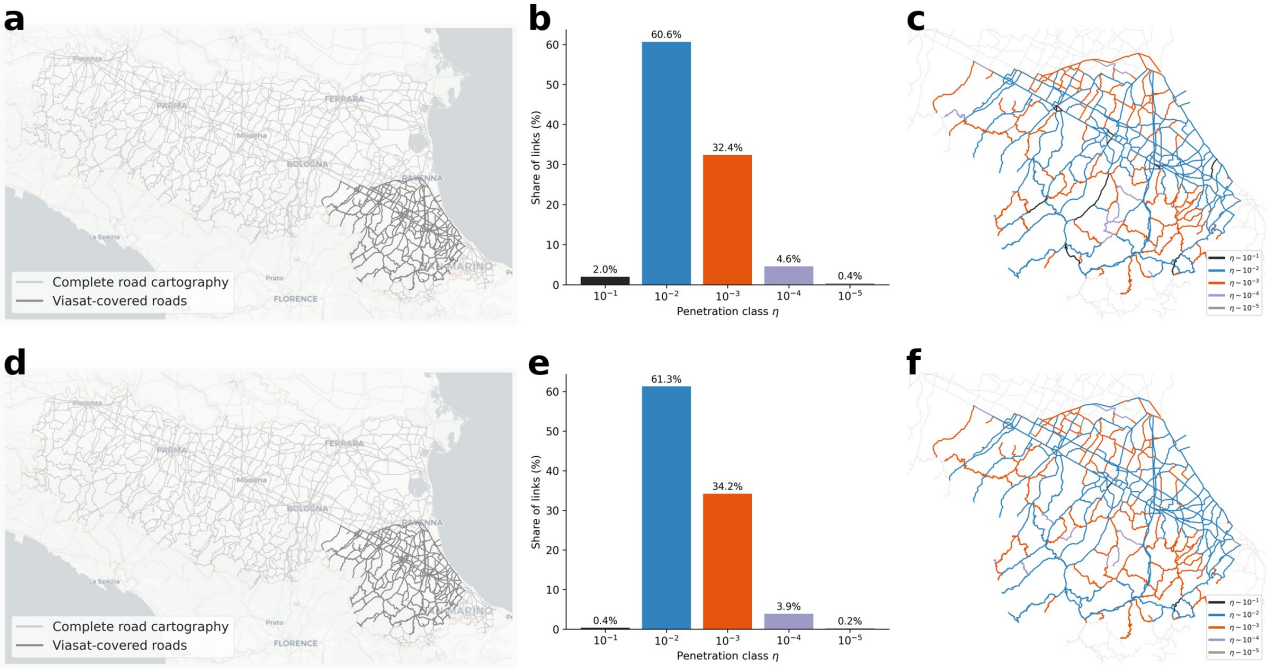


Fig. 7 ~~**Viasat coverage and statistical penetration.**~~ **Viasat coverage and link penetration.** Top row: validation interval 10–17 February 2020. Bottom row: validation interval 28 June–4 July 2021. **(a,d)** Matched Viasat links (dark grey) over the complete regional road graph (light grey). **(b,e)** ~~Shares~~ Share of ~~the principal link-specific~~ usable links in each penetration ~~decades, $\eta \sim 10^{-1}$~~class, ~~$\eta \sim 10^{-2}$ and $\eta \sim 10^{-3}$~~η_i rounded to the nearest power of ten. **(c,f)** Geographical distribution of the same ~~penetration~~ classes in the Forlì–Cesena validation area.

2.5 ~~Temporal subclusters and representative roads~~Profile errors on all matched links

We compute MAE_i on every matched link with non-zero simulated and GPS totals, 848 of the 930 matched links in 2020 and 845 of 927 in 2021. Simulation and GPS cover the same days: eight in 2020 (six weekdays and the weekend of 15–16 February) and seven in 2021 (five weekdays and the weekend of 3–4 July), with weekend days simulated from the weekend OD matrix. Table 1 and Fig. 8 summarize the distributions.

Table 1 Profile error of the assimilated simulation against penetration-normalized Viasat profiles on all usable matched links, by campaign and penetration class η. MAE in percentage points (Eq. (2.6)); ε is the median relative error (Eq. (5)); r is the median Pearson correlation of the hourly shares. Classes with fewer than ten links (three links with $\eta \sim 10^{-5}$ in 2020, two in 2021) are included in the totals only.

Campaign	Class	Links	Median MAE	IQR	10th–90th pct.	ε [%]	r
2020	all	848	1.34	1.05–2.01	0.87–3.28	32	0.82
	10^{-1}	17	0.94	0.91–0.97	0.79–1.12	23	0.91
	10^{-2}	514	1.13	0.96–1.34	0.83–1.61	27	0.87
	10^{-3}	275	2.13	1.66–2.82	1.43–3.65	51	0.66
	10^{-4}	39	5.11	4.63–6.77	3.77–7.50	123	0.36
2021	all	845	1.49	1.15–2.12	0.98–3.48	36	0.76
	10^{-1}	3	1.13	–	–	27	0.85
	10^{-2}	518	1.24	1.03–1.51	0.89–1.86	30	0.83
	10^{-3}	289	2.18	1.73–3.06	1.43–3.89	52	0.64
	10^{-4}	33	5.75	4.86–6.82	4.67–7.64	138	0.27

The ~~most informative penetration stratum is not temporally homogeneous.In~~ median MAE is 1.34 pp in 2020 ~~, the validation archive contains 930 common simulated–Viasat links ,~~ and 1.49 pp in 2021, i.e. a relative error ε of ~~which 827 have usable profiles after filtering; the representative high-penetration class contains~~ 32% and 36% of the mean hourly share. The error depends strongly on penetration. On the 514 links with $\eta \sim 10^{-2}$ the 2020 median is 1.13 pp ($\varepsilon = 27\%$), on the 275 links with $\eta \sim 10^{-3}$ it is 2.13 pp ($\varepsilon = 51\%$), and on the 39 links with $\eta \sim 10^{-4}$ the GPS profile is dominated by counting noise ($\varepsilon = 123\%$). The 2021 values are similar (Table 1).

Part of the error is a systematic lead of the simulation. Summed over all usable links, the simulated profile peaks at 17~~candidate links (Supplementary Section S5). We therefore cluster the hourly simulated profiles of this candidate set using Ward agglomerative clustering. Cutting the dendrogram at a Ward distance of 15,000 yields three subclusters (Fig. 9a). Their spatial distribution is distinct enough to support illustrative validation examples: each cluster contributes one representative road,~~ :00 and the Viasat profile at 18:00 in both campaigns, and the MAE of the two aggregate profiles is 0.69 pp (2020) and 0.82 pp (2021). Shifting the simulated profiles forward by one hour lowers the median link MAE to 1.29 and 1.34 pp. We do not apply this shift, since the simulator output and the Viasat tables both label each hour by its start; the possible origins of the lead are discussed in Section 4.1.

To judge these values we apply two fixed profiles to every link and compute the same error (Table 2). The first is the hourly profile of the OD demand injected in the runs, summed over the same days; it measures what the network loading adds to the timing of the demand. The second is the mean normalized hourly profile of the MTS coils: the ~~selected roads are constrained to remain geographically separated (Fig. 9b). This makes~~

~~the final comparison a test of distinct temporal profiles rather than a repeated inspection of neighbouring links with nearly identical dynamics [38, 39]. The three links are not intended to be the only evidence of validation; the full matched-link statistics, candidate-filtering logic, penetration classes, mean profile discrepancies and selection outcomes are reported in Supplementary Section S5~~82 coil directions with complete data on all eight days in 2020 and, since no observed coil table covers June–July 2021, the summer weekday averages of 471 coil directions from the 2022 archive. The reconstruction has a lower error than the OD profile on 74% of the links in 2020 and 76% in 2021, but the gain is small, from 1.40 to 1.34 pp and from 1.52 to 1.49 pp in median. The MTS mean profile does better than the reconstruction: its median MAE is 1.15 pp in 2020 and 1.21 pp in 2021, and its error is lower on 91% and 87% of the links. On the $\eta \sim 10^{-2}$ class of 2020 the medians are 0.82 pp for the MTS profile and 1.13 pp for the reconstruction. A generic counter profile therefore describes the timing of traffic on the Viasat links better than the simulation, which takes its timing from the OD demand and anticipates the observed peaks by about one hour. What the reconstruction adds is a volume for every directed link and its spatial distribution, which no single profile provides; its hourly shape on unobserved links is its weakest component.

Table 2 Median and interquartile range of the link MAE (pp) for the reconstruction and for two profiles applied identically to every link: the hourly profile of the injected OD demand and the mean normalized profile of the MTS coils (2020: the 82 coil directions with complete data on all eight validation days; 2021: summer weekday averages of 471 coil directions from the 2022 MTS archive). The last column gives the percentage of links on which the assimilated simulation has a lower MAE than the baseline.

Campaign	Profile	Median MAE	IQR	Simulation better [%]
2020	Simulation (assimilated)	1.34	1.05–2.01	–
	Simulation (raw)	1.33	1.02–1.99	–
	Simulation shifted by +1 h	1.29	0.92–1.97	–
	OD demand profile	1.40	1.10–2.03	74
	MTS mean profile	1.15	0.74–1.85	9
2021	Simulation (assimilated)	1.49	1.15–2.12	–
	Simulation (raw)	1.39	1.09–2.05	–
	Simulation shifted by +1 h	1.34	1.04–2.01	–
	OD demand profile	1.52	1.21–2.18	76
	MTS mean profile	1.21	0.84–1.93	13

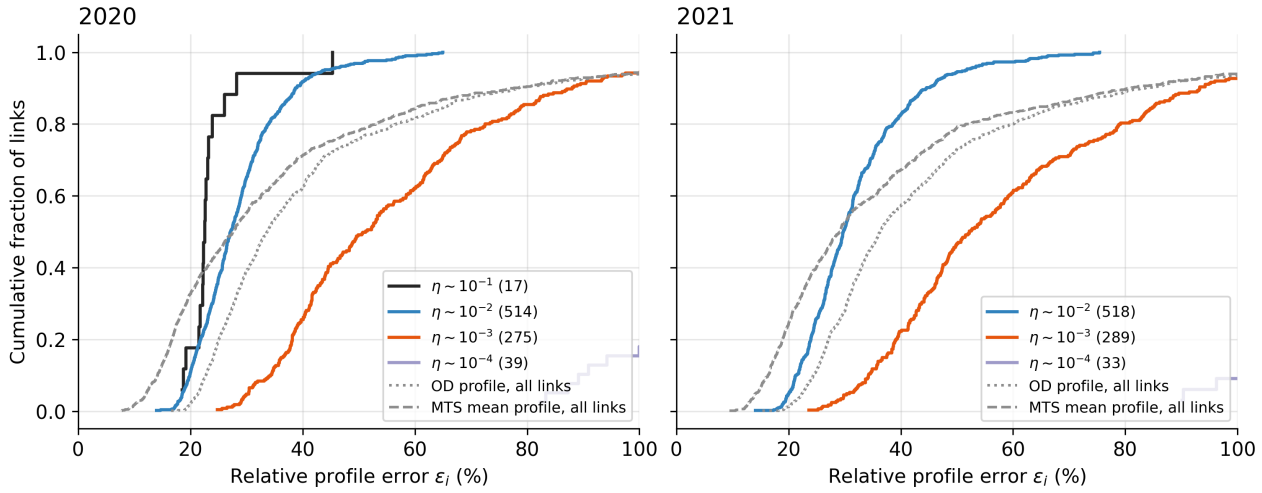


Fig. 8 Profile error on all usable matched links. Cumulative distribution of the relative error ε_i (Eq. (5)) by penetration class, with the number of links in parentheses, for 10–17 February 2020 (left) and 28 June–4 July 2021 (right). Grey curves: the same error when the OD demand profile (dotted) or the MTS mean profile (dashed) is used as prediction on every link.

2.6 Temporal subclusters and representative roads

To show individual comparisons we take the 487 links of the upper class of a three-class k-means partition of $\log_{10} \eta_i$, which corresponds to $\eta_i \geq 0.40\%$ in 2020. Their simulated profiles are normalized to hourly shares and clustered with Ward linkage and Euclidean distance [38, 39]; normalizing removes the volume of each road, so the clusters group links by the shape of their daily profile. Cutting the dendrogram into three clusters (Ward distance 0.15, Fig. 9a) gives subclusters of 51, 78 and 358 links. Subcluster 1 has the sharpest morning peak (7.1% of daily traffic at 07:00) and the fastest evening decline, subcluster 3 the flattest morning and the largest evening tail, and subcluster 2 lies in between (Supplementary Fig. S2). The three subclusters are spread over the whole validation area (Fig. 9b).

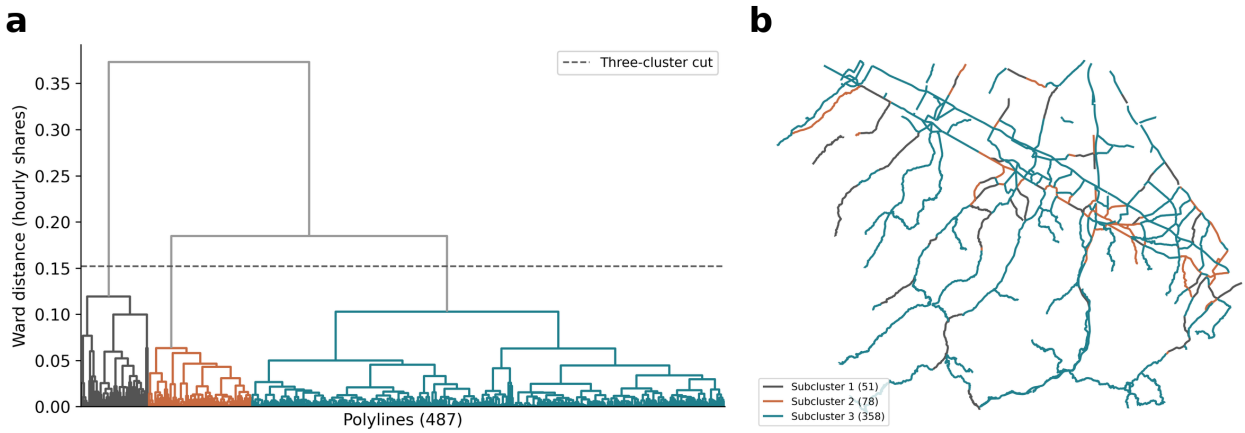


Fig. 9 ~~**Subclustering of the representative GPS-penetration stratum in 2020.**~~ **Subclusters of the high-penetration links in 2020.** **(a)** Ward dendrogram of the normalized hourly ~~link-flow~~ profiles of the 487 candidate links; the dashed line marks the ~~adopted~~ three-cluster cut~~at 15,000.~~ . **(b)** ~~Spatial location~~ Location of the three ~~resulting~~ subclusters, ~~using the same colours as~~ with the ~~dendrogram~~number of links in parentheses.

~~For each selected link , validation compares normalized hourly profiles:~~

$$\epsilon_i(h) = 100 \left| \frac{\phi_i(h)}{\sum_h \phi_i(h)} - \frac{\widetilde{\psi}_i(h)}{\sum_h \widetilde{\psi}_i(h)} \right| .$$

One link per subcluster is shown in Fig. 10. Polyline 104671 is kept ~~as the charcoal representative because it is~~ because it lies close to the ~~urbanized Forlimpopoli area and therefore has a clear geographical interpretation. The remaining two representatives are selected algorithmically from the other clusters by minimizing the mean normalized-profile discrepancy subject to a maximum mean error of 0.80% anda minimum pairwise distance of~~ urban area of Forlimpopoli. The other two are the links of subclusters 2 and 3 with the lowest MAE among those at least 40 km ~~. These thresholds are reporting constraints rather than fitted parameters or optimization targets: the error limit keeps the examples within the sub-percent validation regime, whereas the distance limit prevents selecting three visually redundant neighbouring roads. The resulting links are 187204 and 132443. Their mean discrepancies are 0.53%, 0.76% and 0.61% , respectively, and the minimum pairwise separation is 41.9 km (Fig. 10). The visual agreement of the daily profiles illustrates the reconstructed temporal shape on roads with different cluster membership and different positions in the validation area; broader validation distributions are reported in~~ from the other representatives: polylines 134419 and 64544. They are therefore

best cases. Their MAE is 0.81 and 0.72 pp ($\varepsilon = 19\%$ and 17%), lower than on 95% and 99% of the usable links, while polyline 104671 has MAE = 1.22 pp ($\varepsilon = 29\%$), close to the median of the $\eta \sim 10^{-2}$ class (Supplementary Section S5). On all three roads the simulated morning rise precedes the GPS one by about an hour, which is the lead discussed above.

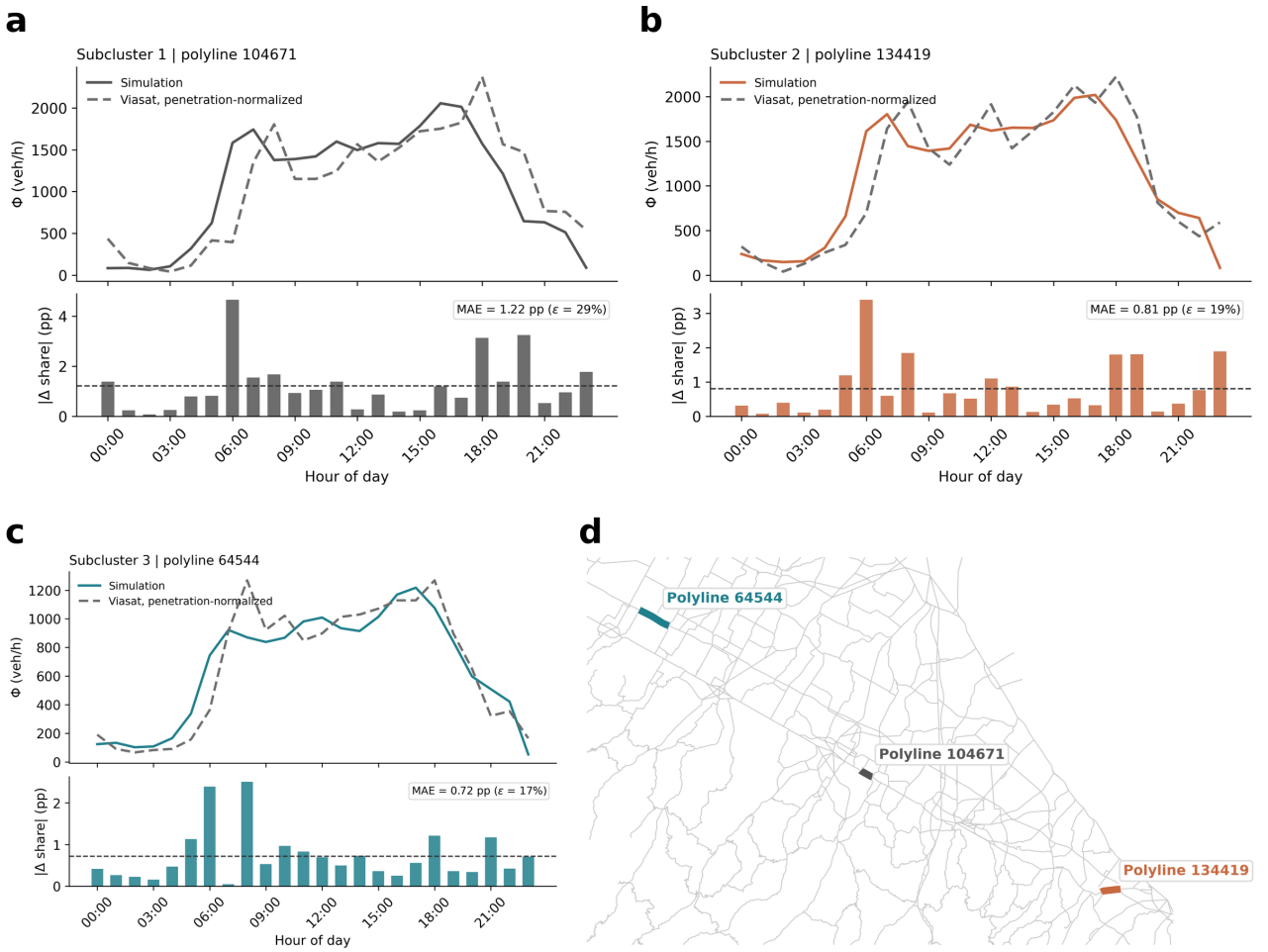


Fig. 10 ~~**Validation on spatially distributed representative polylines, 10–17 February 2020.**~~ **Representative links, 10–17 February 2020.** **(a–c)** Mean simulated hourly flow and ~~incremented~~ penetration-normalized Viasat profile ~~, with normalized-profile discrepancy below~~ (top) and hourly absolute share difference $|p_i(h) - q_i(h)|$ (bottom). **(a)** ~~Polyline~~ Subcluster 1, polyline 104671, ~~mean error 0.53%~~MAE = 1.22 pp. **(b)** ~~Polyline 187204~~Subcluster 2, ~~mean error 0.76%~~polyline 134419, MAE = 0.81 pp. **(c)** ~~Polyline 132443~~Subcluster 3, ~~mean error 0.61%~~polyline 64544, MAE = 0.72 pp. **(d)** ~~CartoDB Positron map~~ Location of the three ~~selected~~ polylines on the road graph, coloured as in Fig. 9.

2.7 Operational webserver

The ~~reconstruction workflow was also implemented as a webserver so that the validated reconstruction chain can be deployed as an operational decision-support service rather than only as an offline analysis script. The dashboard is therefore an interface to the same modelling, data-assimilation and export logic described above, not a separate software contribution. It separates the browser-facing interface from the simulation backend. Through the interface, an authenticated user chooses a simulation interval and submits a run request. The backend generates the corresponding model configuration , launches the~~ same chain runs as a webserver. An authenticated user selects a time interval in the browser; the backend builds the model configuration from the OD tables and the available coil data, runs `minimocas` ~~simulation asynchronously, monitors progress and applies the coil-based data-assimilation step before returning the reconstructed link-flow table . This~~

~~design makes the assumptions, inputs and outputs reproducible while allowing stakeholders to interact with reconstructed traffic fields through maps and tables. It follows the smart-city idea that mobility models become more useful when they are exposed as repeatable analytical services, while still requiring transparent assumptions and validation [2–4, 34].~~

~~This implementation makes three practical outputs available to non-developer users. First, completed simulations are stored with metadata such as~~ in a separate process, applies the coil assimilation and stores the link-by-hour table with the run metadata (user, start time, runtime, ~~completion and status, so previous analyses can be reopened. Second, data-assimilated flows can be exported as a table indexed by road polyline and time. Third, the road network can be visualized interactively: single simulation details combine time-series plots with a map~~ status). Users can download the assimilated flows indexed by polyline and hour, plot the time series of selected links ~~, while aggregate flux maps colour all polylines by reconstructed flow quantiles. The same dashboard includes monitoring views for the MTS coil data , including~~ and map all polylines coloured by flow quantile. The dashboard also shows the MTS data (station maps, time series, rankings, clusters and anomaly diagnostics), so that counter quality and assimilation can be inspected in one place. Supplementary Section S7 describes the ~~webserver components,~~ components, the user workflow and ~~API logic in tabular form~~the API.

3 Methods

3.1 Data sources and preprocessing

The road support is a routable regional graph derived from OpenStreetMap cartography. Road polylines are represented by local and cartographic identifiers, front and tail nodes, length and direction metadata. A preprocessed cartography file defines the node–polyline adjacency used both by the simulator and by the data-assimilation algorithm. The preprocessing follows standard graph-based principles for converting detailed cartography into a transport network that is simple enough to simulate while preserving the topological relations relevant to routing ~~[35, 36, 44? ? –46]~~[35, 36, 45, 46].

Mobile-phone mobility is aggregated to municipality-level OD activity. The January 2020 records are separated into weekdays (Monday–Friday) and weekend days (Saturday–Sunday), with the public holidays of 1 and 6 January counted as weekdays, and into four generation classes: observed source–observed destination (`src-dest`), observed source–random destination (`src-rnd`), random source–observed destination (`rnd-dest`) and random source–random destination (`rnd-rnd`). These classes determine how origins and destinations are sampled when demand is injected into the simulation. ~~The construction keeps the empirical OD component explicit and uses distance-aware random endpoints~~Random endpoints, drawn with a travel-time budget, are used only for residual demand that cannot be assigned to two observed municipalities~~. The random-endpoint classes therefore act as a controlled residual mobility mechanism: they regularize sparse OD mass, preserve coverage of the regional graph and avoid forcing weakly observed flows into deterministic routes with spurious precision. This reflects the sensitivity of~~ , because trip-distribution models are sensitive to spatial support and missing observations ~~[24, 25, 32, 33, 47]~~[24, 25, 32, 33].

MTS coil data are matched to road polylines through barrier metadata. A direction-pivot table maps registry directions to cartographic directions (`IN`/`OUT`). ~~When direct coil observations are missing for the simulation interval, the data-assimilation code fills them using seasonal weekday averages, combining the most recent year and previous years with inverse standard-deviation weights~~In the archived 2020 runs, short gaps in the observed coil series are filled by linear interpolation in time; for the 2021 runs no observed table in the project archive covers the simulated days, and the code falls back to weekday-specific seasonal average profiles. Viasat GPS data are ~~independently~~ map-matched to road polylines ~~and are used only to validate the simulation framework; they are not used to assimilate or tune simulated flows~~independently. The meaning, scale and role of each data layer are summarized in Supplementary Section S1.

3.2 The `minimocas` traffic model

`minimocas` is an agent-based traffic simulator in which each vehicle is represented by an agent called `"pawn"`. A pawn has a current polyline, a position along the polyline, a speed, a destination and a status. Simulation time advances with time step Δt; in the Emilia-Romagna configuration the code supports sub-hourly evolution and hourly camera dumps. During each time step, active pawns move along their current polyline. If they reach a node, the next polyline is chosen from the best-path matrix for the current destination, with an optional probability of missing the best path. The best-path matrix is computed on the weighted graph with a Dijkstra-style engine and respects no-turn constraints. ~~The objective is not behavioural route-choice realism or dynamic congestion assignment. Shortest-path routing, the absence of adaptive reroutingand the simplified congestion rule are deliberate choices that favour stable reconstruction of aggregate directional flow fields from uncertain demand~~Routing is shortest-path without adaptive rerouting, and congestion enters only through the speed rule described below; these choices keep regional runs stable and fast. The model is ~~deliberately~~ simpler than full microscopic platforms such as MATSim and SUMO [41, 42], but it shares their agent-based representation of trips and network loading.

Sources inject new pawns according to time-varying creation rates. The daily creation-rate vector is indexed by simulation time, allowing the mobile-phone hourly demand profile to modulate the number of generated trips. Depending on the generation class, the model can use observed origins and destinations, draw one missing endpoint conditionally, or draw both endpoints from the regional node set. The selected OD pair is converted into an initial directed polyline and a destination node; the pawn then follows the network until the trip is completed or killed by model rules.

The simulator records two forms of traffic output. Barrier counters count crossings at coil-like virtual barriers. Unique polyline output records, for each dump interval and each direction, the number of distinct pawn identifiers present on a polyline. This second output is the link-level quantity used in the reconstructed flow maps and in the subsequent data-assimilation step.

The model also updates link velocities from simulated density. For each polyline i, density is the number of pawns on the link divided by its length. The code imposes a critical-density rule: above the critical density the link speed is set to a critical speed; below it, the nominal polyline speed is linearly reduced with density. The

link flow used internally is the product of density and updated speed. This places the implementation within the broader family of traffic-flow and car-following models that link microscopic motion to macroscopic density, speed and flow [43, 48–55]. Supplementary Section S2 provides the run inventory and source-class counts, while Supplementary Section S3 describes the model-state variables.

3.3 Coil-based data assimilation

The data-assimilation procedure implemented in the framework adjusts the simulated unique-link matrix using MTS coil residuals. Let $u_i(h)$ denote the simulated unique count on the polyline associated with coil i, $b_i(h)$ the corresponding synthetic barrier count, and $c_i(h)$ the observed coil count after direction mapping and temporal harmonization. The code first estimates the relation between barrier and unique counts,

$$r_i(h) = \frac{b_i(h)}{\max(u_i(h), 1)}, \tag{6}$$

and computes a relative residual in barrier space,

$$\Delta_i(h) = \frac{c_i(h) r_i(h) - b_i(h)}{\max(b_i(h), 1)}. \tag{7}$$

~~This residual is converted~~ The residual is turned into an additive ~~assimilation~~ increment on the ~~corresponding directed road count . Negative assimilated counts~~ directed coil link,

$$\delta_i(h) = u_i(h)\,\Delta_i(h), \qquad \tilde{u}_i(h) = u_i(h) + \delta_i(h), \tag{8}$$

which is applied only when $\tilde{u}_i(h) > 0$. Whenever $u_i(h) \geq 1$ and $b_i(h) \geq 1$, Eqs. (6)–(8) give $\tilde{u}_i(h) = c_i(h)$: on the coil link the assimilated count equals the observation. Negative counts produced by the propagation are clipped to zero~~in the final output. The procedure is a local data-assimilation step: fixed counters~~. Fixed counters therefore constrain the simulated field where ~~observations are available, while~~ they exist, and the simulation supplies the spatial structure ~~needed to propagate~~ used to pass this information to nearby ~~unobserved links [28, 31]. It should not be interpreted as a full dynamic traffic-assignment~~ links [28, 31]; this is a local correction, not a calibration of the ~~entire~~ whole network.

The ~~assimilation increment is not applied only at the coil polyline. It is~~ increment is propagated recursively through adjacent directed links ~~using turning weights inferred~~ with turning weights taken from the simulated unique counts. For an incoming ~~directed~~ link (n_1, n_2) and an ~~outgoing neighbour~~ admissible outgoing link (n_2, n), ~~the forward weight is the share of simulated flow continuing toward n among all~~

$$\delta_{(n_2,n)}(h) = w_{(n_1,n_2)\to n}(h)\,\delta_{(n_1,n_2)}(h), \qquad w_{(n_1,n_2)\to n}(h) = \frac{u_{(n_2,n)}(h)}{\sum_m u_{(n_2,m)}(h)}, \tag{9}$$

where the sum runs over the admissible outgoing links. If the outgoing ; when their simulated counts are missing , the weight is replaced by a uniformsplit across admissible alternatives. A symmetric backward recursion the split is uniform. A backward recursion with the analogous upstream weights distributes the same local assimilation increment upstream. The increment to the incoming links. In the runs analysed here the recursion stops when the absolute propagated increment falls below one vehicle, when depth exceeds five after three links, when the next node gives access to another coil polylineis reached, or when a no-turn or patched-closed forbidden turn or a closed polyline would be crossed. The data-assimilated table is written in the same assimilated table keeps the directional unique-polyline format as of the simulator output. The propagation depth, one-vehicle stopping threshold and fallback uniform turning weights are operational parameters used to keep the data assimilation local and stable when simulated turning counts are sparse. Poorly instrumented areas therefore remain more uncertain than roads close to reliable countersAreas far from reliable counters receive no correction and remain as uncertain as the raw simulation. Supplementary Section S4 reports the data-assimilation rules , including the rationale for the one-vehicle stopping threshold and five-link recursion limit, and the coil-282 summary used for the local comparisonlists the stopping rules and the differences between the archived implementation and the one now running in the webserver.

3.4 Accident exposure and hazard-weighted traffic

The road-safety analysis uses regional accident records for 2022–2024 as aggregated counts. The records are suitable for describing temporal distributions and broad road-environment composition, but they are not used as geolocated single-event points. Consequently, the method does not geolocate individual crashes, predict individual accidents or calibrate observed crash hotspots. It combines aggregate accident timing with the simulated traffic field to construct exposure-normalized and hazard-weighted screening indicatorsaccident data are the hourly tables of road accidents with injuries in Emilia-Romagna for 2022, 2023 and 2024, averaged over the three years; they pool all days of the week. Exposure comes from twelve weekday runs, one in each month of 2023, and eleven Sunday runs of the same year, all driven by the OD matrices of Section 3.1.

Let $\phi_i(h)$ be the simulated flow on link i during hour h. Hourly exposure is obtained by summing simulated traffic over the road network,

$$E(h) = \frac{\sum_i \phi_i(h)}{\sum_{h'} \sum_i \phi_i(h')}, \tag{10}$$

so that $E(h)$ is a daily share. Accident counts are normalized in the same way,

$$A(h) = \frac{a(h)}{\sum_{h'} a(h')}, \tag{11}$$

where $a(h)$ is the mean number of accidents in hour h. The exposure-normalized hourly risk is then , and $R(h)$ follows from Eq. (1). Separate weekday and holiday exposure profiles can be used in the denominator when the goal is to compare temporal regimesWeekday and weekend exposure give two versions of $R(h)$; the link indicators below use the weekday one.

~~For spatial screening, the hourly risk profile is projected onto each link through~~ The hazard-weighted exposure ~~,~~ of link i is

$$H_i = \sum_h \phi_i(h) R(h). \tag{12}$$

~~This quantity is high when a link carries large simulated traffic during hours with elevated aggregate relative risk. The weekday–holiday contrast is computed as~~ and the weekday–weekend contrast is

$$D_i = \frac{H_i^{\text{weekday}} - H_i^{\text{holiday}}}{H_i^{\text{weekday}} + H_i^{\text{holiday}}} \frac{H_i^{\text{weekday}} - H_i^{\text{weekend}}}{H_i^{\text{weekday}} + H_i^{\text{weekend}}}, \tag{13}$$

~~with the denominator evaluated only where the total hazard-weighted exposure~~ evaluated where the denominator is positive. ~~Link risk centrality is~~ The risk centrality

$$C_i = \frac{H_i}{\sum_j H_j}, \tag{14}$$

~~and measures~~ is the share of the total hazard-weighted exposure carried by link i. ~~Finally, anomalous hazard relative to exposure is screened by comparing ranks,~~ , and the rank difference

$$S_i = \text{rank}(H_i) - \text{rank}(E_i), \tag{15}$$

~~where~~ compares the percentile rank of H_i with that of the total simulated exposure $E_i = \sum_h \phi_i(h)$ ~~is total simulated link exposure and ranks are oriented so that larger values correspond to higher hazard or exposure. Large positive deviations indicate links whose hazard-weighted exposure rank is high compared with their baseline exposure rank. These indicators support screening and prioritization, but they are not interpreted as proof of causal risk or as geolocation of individual accidents~~among the links with $E_i > 0$. Figure 6e highlights the links in the top 5% of S_i.

3.5 GPS validation and clustering

~~Validation uses Viasat only after matching GPS roads with simulated polylines. The GPS data~~ Viasat data enter only after map matching to the simulated polylines, and are not used to generate demand, assimilate ~~coil residuals~~coils, tune the simulation or ~~select model parameters . They provide an external check of the temporal profiles produced by the reconstruction framework. For every matched road, the link-specific penetration coefficient in Eq~~choose parameters [13, 30, 31]. Eqs. (2)~~rescales the GPS series so that total volume on that road is aligned before profile comparison. Because this scaling uses each road separately, it tests the temporal profile on that physical link rather than a region-wide sum or a universal market-penetration factor. The normalized-profile discrepancy in Eq. (2.6) is used as the hourly validation diagnostic because absolute probe counts are not total traffic counts: Viasat observes a partial fleet, and its penetration varies across links. Normalization isolates whether the simulated and observed profiles rise, peak and decline at comparable hours,~~

which is the validation target for the reconstructed temporal field. This is a conservative use of probe data: the GPS traces validate shape and timing after penetration adjustment, but they are not treated as complete traffic counts and are not merged back into the calibration or data-assimilation stages [13, 30, 31]. (5) define the penetration, the normalized profile and the profile errors; links with zero simulated or GPS total are excluded. For each campaign the simulated days are those of the Viasat extract, so the 5 July 2021 run is not used. The baselines of Table 2 are applied with the same formulas, replacing $p_i(h)$ by the baseline profile.

To choose interpretable examples for the paper, links in the representative penetration class are clustered by their simulated hourly profiles using Ward linkage and Euclidean distance. The dendrogram is cut into three subclusters. Polyline The examples of Fig. 10 are chosen as described in the Results: Ward clustering of the normalized simulated profiles of the high-penetration links, one link per subcluster, with polyline 104671 is fixed as the charcoal representative because of its urban location near Forlimpopoli. Representatives from fixed and the other two subclusters are selected by minimizing mean normalized-profile error while enforcing a maximum mean error of 0.80% and a minimum inter-link separation of selected by lowest MAE under a 40 km . This selection avoids visually redundant neighbouring examples and keeps one validation road for each temporal subcluster. The examples are illustrative representatives of temporal and geographical regimes, not the sole validation evidence. Extended validation tables, threshold rationale, candidate-link counts and representative-link metrics are reported in separation constraint. Supplementary Section S5 lists the selection steps and the metrics of the selected links.

4 Discussion

The main methodological contribution is the construction of an operational reconstruction chain in which heterogeneous mobility sensors keep explicitsemantic roles. Mobile-phone records define workflow keeps the role of each sensor explicit: mobile-phone records give the regional demandsignal; , the road graph and simulator perform the network assignment; coils provide local data assimilation where direct counts exist; the simulator assign it, coils correct the simulated counts where they exist, and Viasat is reserved for independent temporal-profile validation of the simulation framework. Each observational layer preserves its original measurement meaning, and the workflow explicitly separates demand generation, routing, data assimilation and validation. This separation is important because each layer has a different sampling mechanism and a different spatial support. It also follows a central lesson of urban analytics: large kept for validation. Large mobility data sets are powerful most useful when their observation process is kept visible , not when they are treated as complete observations stays visible [1, 3, 4, 34].

The validation results show that the simulation framework can produce reconstructed daily profiles consistent with On all usable matched links the median relative profile error is 32% in 2020 and 36% in 2021, and 27% on the links with $\eta \sim 10^{-2}$. The error falls with penetration, as expected for a probe signal. The baselines set these numbers in context: the reconstruction improves only slightly on the timing of the injected demand and is less accurate than the mean profile of the MTS coils, largely because it anticipates the observed peaks by one hour. The leave-one-out test shows that the assimilation fixes the volume on the coil links and

changes the error on a held-out ~~, incremented GPS profiles on illustrative roads from three temporal subclusters. At the same time, the penetration maps and supplementary validation tables make clear that validation strength is not uniform over the entire cartography. The workflow is therefore best interpreted as a transparent data-fusion and validation pipeline, not as proof that every low-penetration road is reconstructed with the same accuracy. This validation focuses on temporal-profile consistency after penetrationnormalization, rather than on absolute fleet-size estimation. The latter would require additional assumptions and data sources, such as fleet composition, vehicle-kilometres travelled, fuel sales or regional population. The current strategy is more conservative: it tests whether the temporal dynamics produced by the reconstruction framework are consistent with an independent probe-vehicle sample after link-specific normalization, but it does not estimate a universal GPS market share or claim absolute calibration of all links~~coil by about three points. The reconstruction therefore provides a volume for every directed link with a daily shape close to the observed one but no better than a generic counter profile, and its volumes away from coils are not validated by the present data.

The ~~practical value of the workflow is that it produces~~ output is a road-level traffic ~~representation~~ field built from data sources that are ~~normally used separately. This output~~ usually analysed separately. It can be used to ~~identify~~ locate corridors with high ~~reconstructed~~ flows, compare weekday and weekend demand~~scenarios~~, inspect the ~~temporal~~ profile of specific roads, ~~support local traffic-management decisions and provide an exposure layer for downstream analyses such as~~ and provide exposure for emissions, noise ~~,~~ or road-safety ~~screening or infrastructure stress. Because the reconstructed quantity is indexed by road polyline and time~~analyses. Being indexed by polyline and hour, it can ~~also be aggregated after the fact~~ be aggregated to municipalities, provinces, corridors or user-defined areas without ~~re-running the model. This is the type of intermediate mobility layer needed by data-driven planning tools, accessibility studies and urban digital-twin applications [2, 5, 40? ?]~~ new runs. Planning tools and regional digital twins need this kind of intermediate layer [2, 5].

The accident analysis ~~illustrates a downstream use of~~ uses the reconstructed field ~~rather than a second calibration target. The available 2022–2024 accident data are aggregate observations, so the paper does not geolocate individual crashes, interpret the maps as observed crash hotspots or claim to predict individual accidents. Instead, simulated traffic is used as a dynamic exposure denominator. Normalizing accidents by exposure reveals temporal risk regimes that are not visible from accident countsalone: night-time and transition hours can have elevated relative risk even when their absolute counts are moderate, whereas high-volume central hours can have lower relative risk because exposureis also high.~~

~~The observed scaling $A \sim E^{0.72}$ reinforces the same interpretation. Exposure accounts for a large fraction of the temporal accident structure, but the sublinear exponent indicates that accident share does not rise proportionally with simulated traffic share. Road environment, speed~~ as an exposure denominator. With aggregate tables, normalization changes the reading of the hourly counts: the hours around midnight have few accidents but an even smaller simulated exposure, and therefore the highest $R(h)$, while the busy day hours fall below one. The exponent $\alpha = 0.72$ (0.58–0.88) means that across hours the accident share grows less than proportionally with exposure. Speed regime, visibility, ~~behavioural conditions and reporting processes may all contribute to the remaining variation. The spatial hazard-weighted maps should therefore be read as screening~~

~~and prioritization tools: they identify links whose simulated traffic occurs during more hazardous hours, and links whose hazard rank is high relative to their exposure rank, but they do not establish causal mechanisms or event-level accident locations~~road environment and reporting can all contribute [21, 22], and the present data cannot separate them. Because $R(h)$ is regional, the link maps rank roads by the hourly composition of their simulated traffic; geolocated accident records would be needed to test whether the highlighted links are more dangerous.

The webserver ~~implementation is important because it operationalizes the validated reconstruction workflow without changing its scientific assumptions. It~~ turns the offline chain into a repeatable service with the same model and assimilation: a user ~~can choose~~ chooses a time window, ~~launch~~ runs the simulation, ~~monitor progress, download data-assimilated flows and visualize the result on the road graph. This makes the workflow suitable for reproducible scenario analysis, operational reporting and communication with stakeholders who need interpretable traffic fields as maps and tables rather than source code. The dashboard also keeps the MTS monitoring layer close to the simulation layer, allowing counter data quality, anomalies and data-assimilation behaviour to be inspected in the same environment. The detailed service architecture is summarized in Supplementary Section~~ downloads the assimilated flows and maps them, next to the MTS monitoring views (Supplementary Section S7).

4.1 Limitations

The data layers do not cover the same period. The demand is the January 2020 OD matrix in every run. The validation weeks are 10–17 February 2020, a few weeks after the demand period, and 28 June–4 July 2021, a summer week one and a half years later; the accident tables cover 2022–2024 and the exposure runs are days of 2023. All runs therefore use a winter, pre-pandemic demand, including those for summer 2021 and for 2023. The 2021 errors are only slightly larger than the 2020 ones (median MAE 1.49 against 1.34 pp), so the daily shape of the demand seems to change less than its level, but the present data cannot test volumes. Within each validation week the composition of days is the same on both sides: weekend days are simulated with the weekend OD matrix, and the archived 2021 set contained one extra Monday run (5 July), a copy of the 28 June configuration, which we excluded because the Viasat extract ends on 4 July. The accident tables pool all days of the week, while the link indicators use weekday exposure.

The ~~main limitation of the present implementation is the seasonal support of the OD matrix. The demand layer used here is based on winter mobility observations, and therefore represents a specific seasonal configuration of regional mobility. Access to multiple OD matrices, ideally derived from different data sources and covering different seasons, would allow the same framework to reproduce seasonal changes in mobility demand more explicitly and to test how robust the reconstructed link-flow field is under different demand regimes.~~ timing of the simulated profiles has two known defects. The simulated peaks precede the Viasat ones by about one hour (Section 2.5). Both series label each hour by its start, and we could not identify the cause from the archived inputs; candidates are the time convention of the hourly OD aggregation and the injection of all trips at their origin without a spread of departure times. In addition, the configuration reads only 23 of the

24 hourly OD values, so no trips are generated between 23:00 and 24:00 and every simulated profile drops in the last hour of the day. Both defects weigh most on the night hours, where $R(h)$ reaches its largest values: at 00:00 the simulated share of daily traffic on the usable Viasat links is 0.65% against 1.05% for Viasat in 2020.

The MTS tables store UTC timestamps, and the assimilation code compares them with the local clock of the simulation without conversion, so in February 2020 each coil hour is imposed one hour early. Rerunning the 2020 assimilation with the coil timestamps converted to local time lowers the median MAE on the Viasat links from 1.34 to 1.31 pp; the figures and tables of this paper use the archived assimilation.

~~A second limitation is methodological and operational.~~ The simulator is ~~intentionally simplified to keep regional runs~~ deliberately simple, so that regional runs stay compatible with webserver ~~deployment and interactive stakeholder~~ use. It ~~does not implement a full~~ has no dynamic congestion mechanism ~~or adaptive traffic assignment; instead, it uses shortest-path network loading together~~ and no adaptive assignment: trips follow shortest paths with a free-flow ~~/thresholdvelocity representation. This choice is coherent with the main target of the study, which is the reconstruction of~~ speed reduced above a density threshold. This is adequate for aggregate directional flows on a large extra-urban ~~road network rather than the detailed modelling of~~ network and inadequate for urban congestion waves. ~~Within this setting, the coil-based data-assimilation step is the mechanism that anchors simulated flows to local observations while preserving feasible runtimes for repeated operational analyses~~A demand basis with several OD matrices, covering different seasons and possibly different sources, would allow seasonal changes to be reproduced explicitly and the reconstruction to be tested under different demand regimes.

Future work should ~~therefore focus on extending the demand basis and testing the model under richer seasonal and multi-source OD scenarios~~first correct the timing defects and the coil clock, then extend the demand basis. On the operational side, the webserver could be extended with explicit scenario controls, seasonal demand selection, user-defined regions of interest and automatic reporting of the OD and sensor layers used in each run.

5 Conclusions

~~This work presents a regional-scale workflow for reconstructing road traffic flows by combining~~ We reconstructed hourly traffic flows on the regional road network of Emilia-Romagna from mobile-phone ~~activity, fixed traffic counters~~OD activity, inductive loop detectors, OpenStreetMap cartography and ~~GPS telemetry. The contribution is not a new routing algorithm, but an interpretable multi-layer reconstruction framework in which~~ Viasat GPS telemetry, keeping demand generation, network loading, local ~~data assimilation and independent validation remain explicitly separated. The proposed chain integrates heterogeneous mobility observations into a coherent time-dependent traffic field at the road-link scale over a large regional network~~assimilation and validation as separate steps.

~~Held-out Viasattelemetry validates the simulation framework by showing strong consistency between reconstructed and GPS-derived hourly profiles on representative roads from different temporal mobility regimes after link-specific penetration normalization. This should be interpreted as a validation of reconstructed~~

temporal dynamics rather than as a claim of complete absolute calibration across the entire network. The results nevertheless show that large-scale mobility patterns can be reconstructed with sufficient temporal and spatial coherence to support downstream operational analyses. Against Viasat, the reconstructed hourly profiles have a median relative error of 32% (2020) and 36% (2021) on all usable links, and 27% where the probe penetration is near 1%. The simulation leads the observed peaks by one hour, gains little over the timing of the demand itself and is less accurate than a mean coil profile; the coil assimilation corrects the instrumented links and has a small effect on the others. These results point to the parts of the chain to improve first: the time convention of the demand, the clock of the coil data and the reach of the assimilation.

A central result of the study is that the reconstructed traffic field can also be reused as a dynamic The same field can serve as an exposure layer for regional road-safety assessment. By combining simulated road exposure with aggregate accident statistics, the framework makes it possible to distinguish highly trafficked corridors from road segments exhibiting disproportionately high hazard relative to their estimated exposure. The analysis reveals marked temporal risk heterogeneity across the day and highlights the existence of road segments whose accident intensity cannot be explained solely by traffic volume. This transforms the reconstructed mobility field into a practical screening tool for identifying potentially critical infrastructures and prioritizing monitoring or intervention strategies. Normalized by simulated exposure, the hourly accident share of 2022–2024 gives a relative risk that peaks around midnight and stays below one after 19:00, and across hours the accident share scales as $E^{0.72}$. The link maps derived from this regional profile rank roads by the hourly composition of their traffic, and can indicate where geolocated accident data or local counts should be collected first.

Beyond scientific analysis, the workflow was designed with stakeholder-oriented applications in mind. The dashboard implementation supports simulation management, monitoring, visualization and export of reconstructed flows , enabling regional analysts to inspect mobility scenarios without directly interacting with the modelling pipeline. Potential applications include congestion screening, infrastructure stress assessment, environmental-impact analysis, exposure-based safety prioritization and digital-twin support for regional mobility management.

Future developments should incorporate uncertainty quantification, systematic sensitivity analyses for the data-assimilation procedure and broader validation campaigns across additional territories and temporal conditions. Further extensions could integrate weather conditions, road characteristics and near-real-time mobility observations to support predictive hazard assessment and adaptive traffic-management strategiesThe webserver delivers the assimilated link flows to regional analysts who do not run the modelling chain. Further work should add uncertainty estimates for the link volumes, a demand basis covering several seasons, and validation on other territories with geolocated accident data and denser counter networks.

Supplementary Material

Supplementary Material provides tabular documentation of The Supplementary Material documents the data layers, simulation runs, `minimocas` state variables, data-assimilation rules , validation metricsassimilation

rules and leave-one-out test, validation tables, accident descriptors and webserver architecturereferenced throughout the main text. Supplementary Fig. S1 reports additional descriptive statistics on accident totals, road environment and seasonal shares, and Supplementary Section S8 summarizes reproducibility notesfor the reported counts, thresholds and operational workflow, and gives reproducibility notes.

Availability of data and materials

Processed figure inputs and the code required to regenerate the analyses are available in the project repository. The manuscript sources, the analysis scripts and the aggregate result tables are available at https://github.com/federicobellisardi/multi-sensor-traffic-reconstruction. The `minimocas` simulator is available at https://github.com/physycom/minimocas. **[TODO: check that both repositories are public before submission]** Public traffic-counter data from the Emilia-Romagna Region are available through the MTS online mobility platform at https://mobilita.regione.emilia-romagna.it/servizi/flussi-online. Road-accident statistics are publicly available from ISTAT at https://www.istat.it/tag/incidenti-stradali/. Mobile-phone and Viasat GPS datasets used in this study are proprietary and cannot be publicly distributed.

Competing interests

Ethics approval and consent to participate

The authors declare that they have mobile-phone data were provided by the operator already anonymized and aggregated, as hourly counts of devices moving between census areas (ACE), which we further aggregated to municipalities. No individual-level data were accessed or processed, and the data were handled in compliance with the EU General Data Protection Regulation (GDPR). **[TODO: confirm with the data provider the aggregation level of the delivered data (ACE or municipality** Consent to participate was not applicable.

Competing interests

C.M. and A.F. are employed by Mindicity, which develops the operational platform described in this work. The other authors declare no competing interests.

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Authors' contributions

FB conceived part of the original study design, developed the reconstruction workflow, performed the analyses and prepared the first manuscript draft. CM and AF contributed to the development of the modelling framework. AB contributed to the study design, interpretation of the modelling results and manuscript revision. All authors read and approved the final manuscript.

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